



Teyzix Core Internship (June Batch)

Task Information

- **Task Title:** Customer Churn Prediction System
- **Task ID:** ML-2
- **Domain:** Machine Learning
- **Assigned Date:** 20th June 2026
- **Submission Deadline:** 29th June 2026

Objective

Develop a Machine Learning solution capable of predicting whether a customer is likely to stop using a company's service based on historical customer behavior and engagement patterns.

Problem Statement

Customer retention is one of the biggest challenges faced by subscription-based businesses, telecom companies, SaaS platforms, banks, and e-commerce businesses. Losing customers directly impacts revenue and business growth.

Your task is to build a machine learning system that analyzes customer data, identifies patterns associated with customer churn, and predicts which customers are at risk of leaving.

Task Requirements

Dataset Requirement

Interns must create their own realistic dataset.

The dataset should contain a minimum of **1,500 records** and include customer-related information.

The dataset must contain relevant attributes such as:

- Customer ID
- Age
- Gender
- City
- Subscription Type
- Monthly Spending
- Tenure
- Number of Purchases
- Customer Support Requests
- Login Frequency
- Last Activity Date
- Satisfaction Score
- Churn Status (Target Variable)

Additional features may be added if required.

Data Preparation

- Handle missing values
- Remove duplicates
- Encode categorical variables
- Feature engineering
- Data normalization or scaling (if required)
- Train-test split

Exploratory Data Analysis

Analyze:

- Customer demographics
- Spending patterns
- Churn distribution
- Feature relationships
- Correlation analysis

- Customer behavior trends

Machine Learning Model

Develop one or more predictive models capable of classifying customer churn.

Possible approaches include:

- Decision Trees
- Random Forest
- Logistic Regression
- XGBoost
- Support Vector Machines
- Neural Networks
- Any other suitable technique

Model Evaluation

Evaluate model performance using appropriate metrics such as:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC
- Confusion Matrix

Prediction Interface

Create a simple interface that allows users to:

- Input customer details
- Predict churn probability
- Display prediction results
- Show important contributing factors

The interface may be:

- Web Application
- Desktop Application
- Notebook-based Interface
- Command-Line Application

Reporting

Prepare a report including:

- Dataset Description
- Data Preparation Process
- Feature Engineering
- Model Selection
- Evaluation Results
- Key Findings
- Business Recommendations

Technology Flexibility

Interns are free to use any machine learning framework, programming language, database, visualization tool, deployment approach, or methodology of their choice.

Examples include but are not limited to:

- Python
- Scikit-learn
- TensorFlow
- PyTorch
- XGBoost
- LightGBM
- Pandas
- NumPy
- Jupyter Notebook
- Streamlit
- Flask
- FastAPI

The choice of technology will not affect evaluation. The primary focus will be on data preparation, model development, evaluation methodology, problem-solving approach, and overall implementation quality.

Bonus Features (Optional)

- Explainable AI (XAI)
- Feature Importance Analysis
- Hyperparameter Optimization
- Customer Segmentation
- Churn Risk Dashboard
- Multiple Model Comparison

- Model Deployment

Deliverables

- Self-Created Dataset
- Source Code
- Trained Model Files
- README Documentation
- Analysis Report
- Screenshots or Demo Video
- Deployment Link (Optional)
- GitHub Repository

Evaluation Criteria

- Dataset Quality and Realism – 15%
- Data Preparation and Feature Engineering – 20%
- Model Performance and Evaluation – 30%
- Problem-Solving Approach – 20%
- Documentation and Presentation – 15%

Difficulty Level

Intermediate

Industry Relevance

Customer churn prediction is one of the most widely used machine learning applications in modern businesses. Telecom companies, SaaS platforms, banks, insurance providers, subscription services, and e-commerce businesses use churn prediction systems to proactively retain customers and improve revenue. This project reflects a real-world machine learning workflow involving data generation, feature engineering, predictive modeling, evaluation, and business-driven decision making.